

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5(a)(i)  | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 wider / thicker / larger diameter → increases / higher, speed ;</li> <li>2 large(r) membrane surface area so more ion movement ;</li> <li>3 myelin → increases / faster / higher, speed ;</li> <li>4 saltatory conduction (in myelinated) ;</li> <li>5 pairwise data comparison ;</li> </ol>                                                                                                                                                                                                                  | 3 |
| 5(a)(ii) | <p>any <b>one</b> from:</p> <ol style="list-style-type: none"> <li>1 impulses, for touch / in <b>A<sup>β</sup></b> neurones, travel faster ;</li> <li>2 (<b>A<sup>β</sup></b> neurones) inhibit, spinal cord / relay / projection, neurones ;</li> <li>3 AVP ;</li> </ol>                                                                                                                                                                                                                                                                                                            | 1 |
| 5(b)(i)  | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 endorphins bind to (endorphin) receptors ;</li> <li>2 stop, calcium ions / Ca<sup>2+</sup>, entering presynaptic knob ;</li> <li>3 no / fewer, vesicles, move towards / fuse with, (presynaptic) membrane ;</li> <li>4 no / less, <u>acetylcholine / ACh</u>, released ;</li> <li>5 no / less, binding of ACh to postsynaptic receptors ;</li> <li>6 no / less, depolarisation of postsynaptic, neurone / membrane ;</li> <li>7 no / fewer, action potentials / impulses, to, pain centre / brain ;</li> </ol> | 4 |
| 5(b)(ii) | <p>any <b>one</b> from:</p> <ol style="list-style-type: none"> <li>1 (drug) dependency / addiction / tolerance ;</li> <li>2 (drug) side-effects ;</li> <li>3 expensive ;</li> <li>4 work more slowly ;</li> </ol>                                                                                                                                                                                                                                                                                                                                                                    | 1 |